

Practice Intermediate Algebra Test III (Version 1)

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Intermediate Algebra).

II) Show all work.

III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

IV) You will need a calculator for this test.

V) You do not have to write the Roman numeral for each exponent rule that you use on this test.

VI) Do not leave your answers with fraction exponents, negative exponents, zero power exponents, or 1st power exponents.

VII) Each problem is worth four points for a total of 100 points.

Simplify

$$\textcircled{1} \left(\frac{7a^3y}{b^4} \right)^0$$

$\boxed{1}$

$$\textcircled{2} 1000^{-\frac{1}{3}}$$

$$\frac{1}{1000^{\frac{1}{3}}}$$

$$\frac{1}{\sqrt[3]{1000}}$$

$$\boxed{\frac{1}{10}}$$

$$\textcircled{3} \sqrt[4]{16^2 \cdot 16^3} = \sqrt[4]{16^5} = (\sqrt[4]{16})^5 = 2^5 = \boxed{32}$$

$$\textcircled{4} \frac{1}{9^{-\frac{5}{2}}} = 9^{\frac{5}{2}} = \sqrt{9^5} = (\sqrt{9})^5 = 3^5 = \boxed{243}$$

$$\textcircled{5} \left(\frac{x \cdot x^2}{x^4} \right)^{-3}$$

$$\left(\frac{x^3}{x^4} \right)^{-3}$$

$$(x^{-1})^{-3}$$

$$\boxed{x^3}$$

$$\textcircled{6} \left(\frac{x^{\frac{1}{3}}}{y^{\frac{2}{3}}} \right)^3$$

$$\frac{(x^{\frac{1}{3}})^3}{(y^{\frac{2}{3}})^3}$$

$$\frac{x^1}{y^2}$$

$$\boxed{\frac{x}{y^2}}$$

⑦ Convert the logarithm to a sum or difference of multiples of logarithms.

$$\log_5(3x^2y^4) = \log_5 3 + \log_5 x^2 + \log_5 y^4$$

$$= \boxed{\log_5 3 + 2\log_5 x + 4\log_5 y}$$

Solve.

$$\textcircled{8} \quad 7^{\frac{x}{2}-1} = 7$$

$$7^{\frac{x}{2}-1} = 7^1$$

$$\frac{x}{2}-1 = 1$$

$$\boxed{x=4}$$

$$\textcircled{10} \quad \log_3 X = -2$$

$$3^{-2} = X$$

$$\frac{1}{3^2} = X$$

$$\boxed{\frac{1}{9} = X}$$

$$\textcircled{9} \quad 9^{2x} = 27^{x+1}$$

$$(3^2)^{2x} = (3^3)^{x+1}$$

$$3^{4x} = 3^{3x+3}$$

$$4x = 3x+3$$

$$\boxed{x=3}$$

$\textcircled{11}$ Find the value of $\log_6 36$ without a calculator.

$$x = \log_6 36$$

$$6^x = 36$$

$$6^x = 6^2$$

$$\boxed{x=2}$$

Solve. If an answer is irrational, round to the nearest ten thousandth.

$$\textcircled{12} \quad \frac{\sqrt[3]{2}}{2} = \frac{8^x}{4^x}$$

$$\frac{2^{\frac{1}{3}}}{2^1} = \frac{(2^3)^x}{(2^2)^x}$$

$$2^{-\frac{2}{3}} = \frac{2^{3x}}{2^{2x}}$$

$$2^{-\frac{2}{3}} = 2^x$$

$$\boxed{-\frac{2}{3} = x}$$

$$\textcircled{13} \quad 0 = 5^{2x} - 4(5^x) + 3$$

$$0 = (5^x)^2 - 4(5^x) + 3$$

$$\boxed{y = 5^x}$$

$$0 = (y)^2 - 4(y) + 3$$

$$0 = y^2 - 4y + 3$$

$$0 = (y-3)(y-1)$$

$$\swarrow \quad \searrow$$
$$y=3 \quad \text{or} \quad y=1$$

$$5^x = 3 \quad 5^x = 1$$

$$x = \log_5 3 \quad 5^x = 5^0$$

$$\boxed{x \approx 0.6826} \quad \boxed{x=0}$$

$$\textcircled{14} \text{ Simplify: } (a^3 \cdot b^2 \cdot a^4)^{-2} = (a^7 \cdot b^2)^{-2} = (a^7)^{-2} \cdot (b^2)^{-2}$$

$$= a^{-14} \cdot b^{-4}$$

$$= \boxed{\frac{1}{a^{14} b^4}}$$

Solve.

$$\textcircled{15} \quad 3(5^{2x^2-x}) - 8 = 367$$

$$3(5^{2x^2-x}) = 375$$

$$5^{2x^2-x} = 125$$

$$5^{2x^2-x} = 5^3$$

$$2x^2 - x = 3$$

$$2x^2 - x - 3 = 0$$

$$(2x-3)(x+1) = 0$$

$$\swarrow \quad \searrow$$
$$2x-3=0 \quad \text{or} \quad x+1=0$$

$$\boxed{x = \frac{3}{2} \quad \text{or} \quad x = -1}$$

$$\textcircled{16} \quad \frac{1}{36^{-x}} = \sqrt{216}$$

$$36^x = 216^{\frac{1}{2}}$$

$$(6^2)^x = (6^3)^{\frac{1}{2}}$$

$$6^{2x} = 6^{\frac{3}{2}}$$

$$2x = \frac{3}{2}$$

$$\boxed{x = \frac{3}{4}}$$

$\textcircled{17}$ If $\log_k 4 \approx 1.26$ and $\log_k 5 \approx 1.46$, use this information to approximate the following value.

$$\log_k 20 = \log_k (4 \cdot 5) = \log_k 4 + \log_k 5$$

$$\approx 1.26 + 1.46$$

$$\approx \boxed{2.72}$$

Solve.

$$\textcircled{18} \quad 5 = 4 + \frac{\log_2(3x-5)}{4}$$

$$1 = \frac{\log_2(3x-5)}{4}$$

$$4 = \log_2(3x-5)$$

$$2^4 = 3x-5$$

$$16 = 3x-5$$

$$21 = 3x$$

$$\boxed{x = 7}$$

$$\textcircled{19} \quad 0 = \log_4 7 - \log_4 6 + \log_4 x$$

$$0 = \log_4 \left(\frac{7}{6}\right) + \log_4 x$$

$$0 = \log_4 \left(\frac{7}{6}x\right)$$

$$4^0 = \frac{7}{6}x$$

$$1 = \frac{7}{6}x$$

$$6 = 7x$$

$$\boxed{\frac{6}{7} = x}$$

Write your answer as an exact expression.

$$(20) \log_5 (X-7) + \log_5 X = 1$$

$$\log_5 [(X-7)X] = 1$$

$$\log_5 (x^2 - 7x) = 1$$

$$5^1 = x^2 - 7x$$

$$5 = x^2 - 7x$$

$$0 = x^2 - 7x - 5$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(1)(-5)}}{2(1)}$$

$$x = \frac{7 \pm \sqrt{49 + 20}}{2}$$

$$x = \frac{7 \pm \sqrt{69}}{2}$$

$$\boxed{x = \frac{7 + \sqrt{69}}{2}} \text{ or } x \neq \frac{7 - \sqrt{69}}{2}$$

$$(21) \log_{(x-2)} (6x^2 + 14x + 13) = 2$$

$$(x-2)^2 = 6x^2 + 14x + 13$$

$$(x-2)(x-2) = 6x^2 + 14x + 13$$

$$x^2 - 2x - 2x + 4 = 6x^2 + 14x + 13$$

$$x^2 - 4x + 4 = 6x^2 + 14x + 13$$

$$0 = 5x^2 + 18x + 9$$

$$0 = (5x+3)(x+3)$$

$$5x+3=0$$

$$\text{or}$$

$$x+3=0$$

$$x = -\frac{3}{5}$$

$$x = -3$$

No Solution

Write your answer as an exact expression.

$$(22) \log_8 3 - 2 \log_8 (3x) = 2 \log_8 5$$

$$\log_8 3 - \log_8 (3x)^2 = \log_8 5^2$$

$$\log_8 \left[\frac{3}{(3x)^2} \right] = \log_8 25$$

$$\frac{3}{(3x)^2} = 25$$

$$\frac{3}{9x^2} = 25$$

$$\frac{1}{3x^2} = 25$$

$$1 = 75x^2$$

$$\frac{1}{75} = x^2$$

$$\pm \sqrt{\frac{1}{75}} = x$$

$$x \neq -\sqrt{\frac{1}{75}} \text{ or } x = \sqrt{\frac{1}{75}}$$

$$x = \sqrt{\frac{1}{75}}$$

$$x = \frac{\sqrt{1}}{\sqrt{75}}$$

$$x = \frac{1}{5\sqrt{3}}$$

$$x = \frac{\sqrt{3}}{5 \cdot 3}$$

$$= \boxed{\frac{\sqrt{3}}{15}}$$

$$(23) \frac{\log_7 X + 5}{1 + \log_7 X} = 3$$

$$\log_7 X + 5 = 3(1 + \log_7 X)$$

$$\log_7 X + 5 = 3 + 3 \log_7 X$$

$$5 = 3 + 2 \log_7 X$$

$$2 = 2 \log_7 X$$

$$1 = \log_7 X$$

$$7^1 = X$$

$$\boxed{7 = X}$$

Simplify.

$$\textcircled{24} \frac{(3xy^{-4})^2}{(x^2y^5)^{-1}} \cdot \left(\frac{x^{-5}y^0}{2b^6}\right)^2$$

$$\frac{3^2 \cdot x^2 \cdot (y^{-4})^2}{(x^2)^{-1} \cdot (y^5)^{-1}} \cdot \left(\frac{x^{-5}(1)}{2b^6}\right)^2$$

$$\frac{9x^2y^{-8}}{x^{-2}y^{-5}} \cdot \frac{(x^{-5})^2}{(2b^6)^2}$$

$$9x^4y^{-3} \cdot \frac{x^{-10}}{2^2(b^6)^2}$$

$$\frac{9x^{-6}y^{-3}}{4b^{12}}$$

$$\boxed{\frac{9}{4x^6y^3b^{12}}}$$

$$\textcircled{25} \left(\frac{5}{x^6} \cdot \frac{x^8}{y} \div \frac{b}{2y^{-1}}\right)^2$$

$$\left(\frac{5x^8}{x^6y} \div \frac{b}{2y^{-1}}\right)^2$$

$$\left(\frac{5x^2}{y} \cdot \frac{2y^{-1}}{b}\right)^2$$

$$\left(\frac{10x^2y^{-1}}{yb}\right)^2$$

$$\left(\frac{10x^2y^{-2}}{b}\right)^2$$

$$\frac{(10x^2y^{-2})^2}{b^2}$$

$$\frac{10^2 \cdot (x^2)^2 \cdot (y^{-2})^2}{b^2} = \frac{100x^4y^{-4}}{b^2} = \boxed{\frac{100x^4}{y^4b^2}}$$